

ECE 447

Lesson 18

Signal sampling and reconstruction

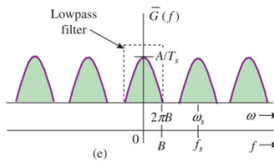
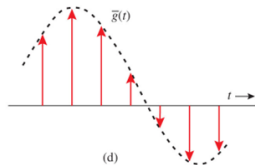
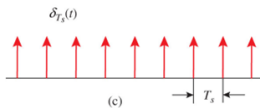
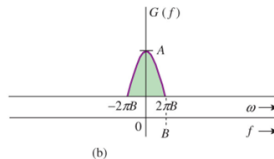
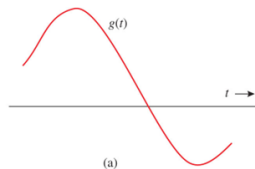


UNITED STATES
AIR FORCE
ACADEMY

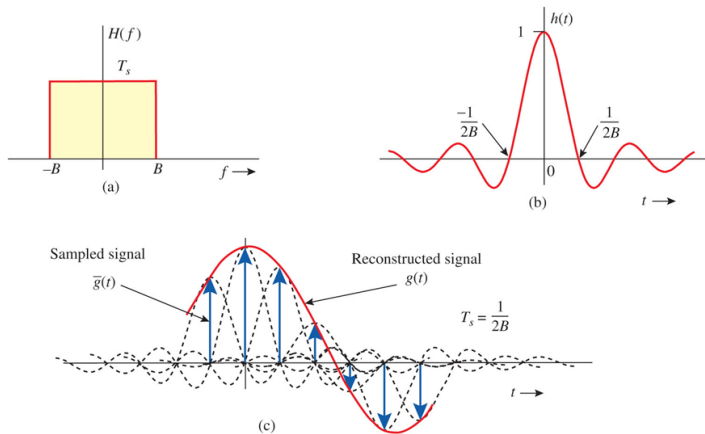
SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **HW4.** Assigned today. Due Lesson 21 (Tue, 6 Oct) to Gradescope.
 - **Lab 4.** Due Lesson 19 (Wed, 30 Sep) to Gradescope.
 - Everything from today through Lesson 32 is GR2 material (Chapters 5-8). GR2 is Lesson 35 (Tue, 17 Nov).

SAMPLING

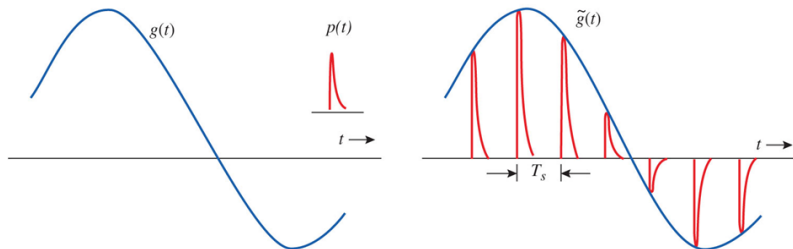


IDEAL RECONSTRUCTION



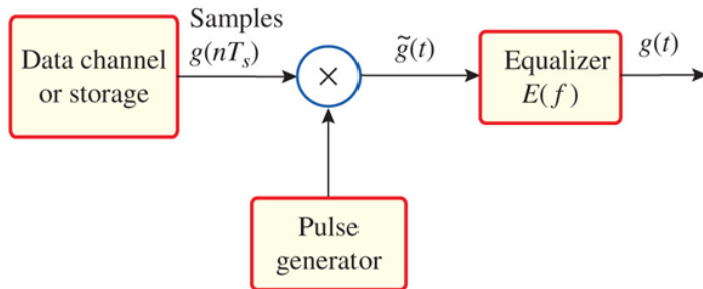
NONIDEAL RECONSTRUCTION (INTERPOLATION)

- Interpolation pulse $p(t)$
- $\tilde{g}(t) = p(t) * \bar{g}(t)$
- $\tilde{G}(f) = P(f)\bar{G}(f) = P(f)\frac{1}{T_s} \sum_n G(f - nf_s)$



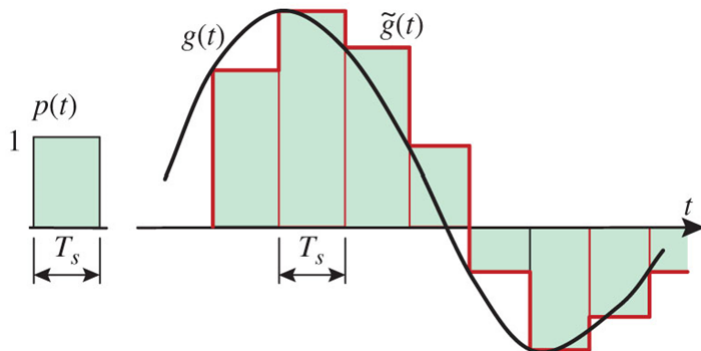
EQUALIZERS

- $P(f)$ filters sampled signal - need to reverse filtering to obtain original
- Use equalizers!



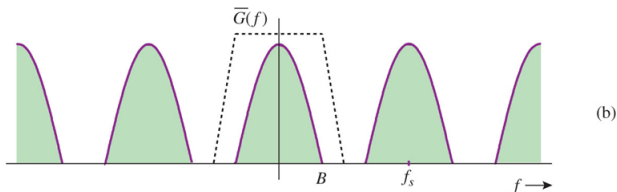
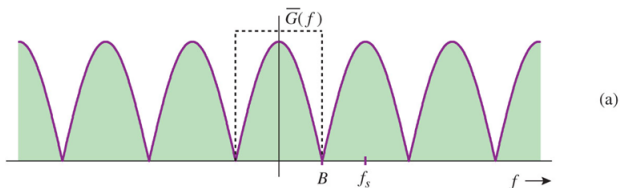
INTERPOLATOR EXAMPLE - FLAT TOP PULSES

- Also known as a zero-order-hold pulse
- Samples signal and holds that value for T_s

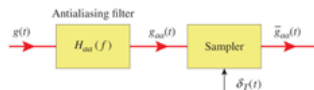
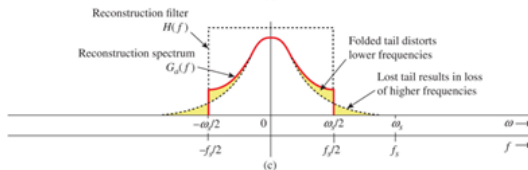
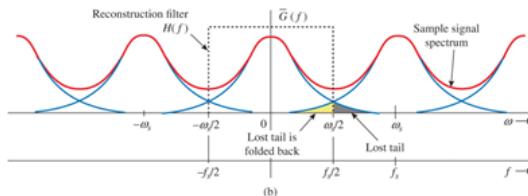
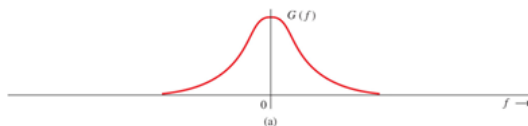


REALIZABLE RECONSTRUCTION FILTERS

- Can't use ideal LPFs
- Need space between samples in spectra



ALIASING



EXAMPLE PROBLEM - 5.1-7